

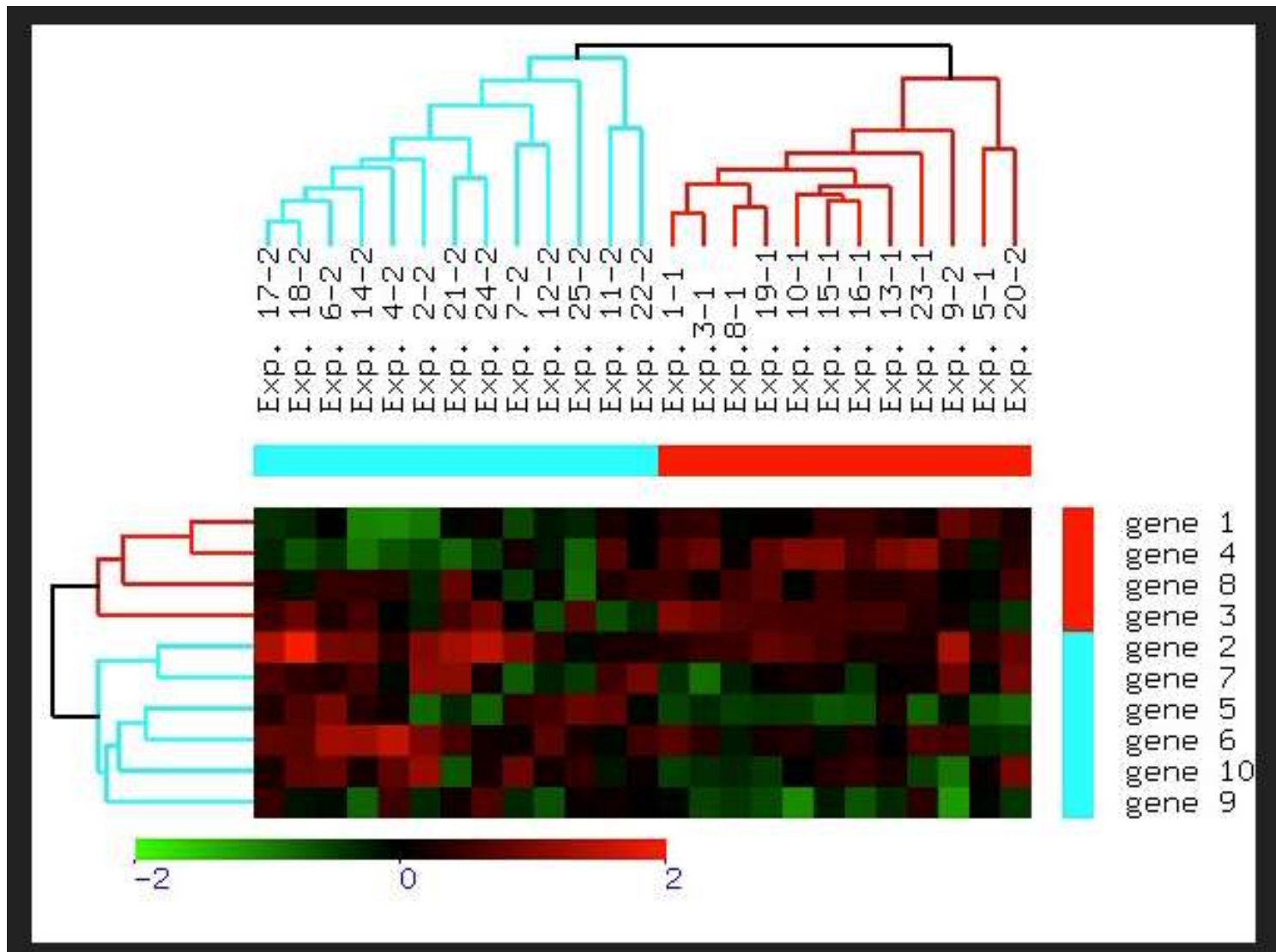
Clustering IV: Hierarchical Clustering

Statistical Data Mining II

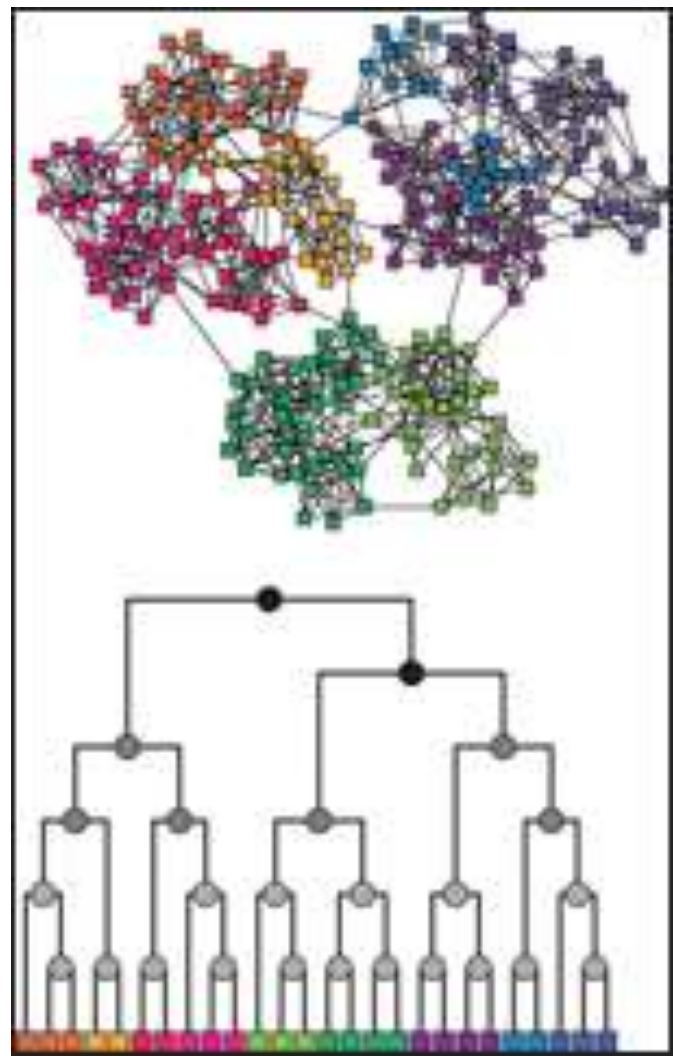
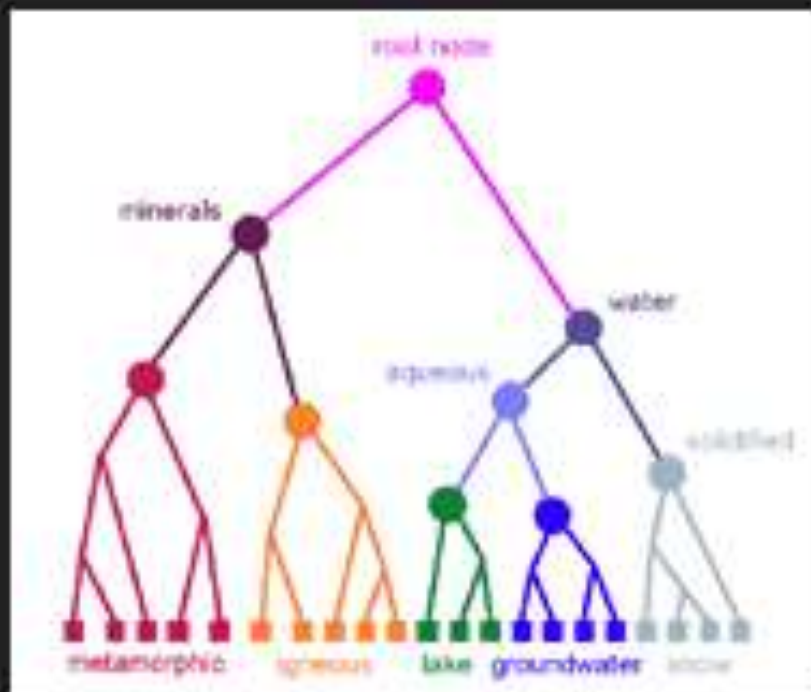
Rachael Hageman Blair

Outline

- Introduction
- Dendrogram
- When to Stop?
- Agglomerative Clustering
- Different types of linkage
- Example
- Computation



Clustering IV



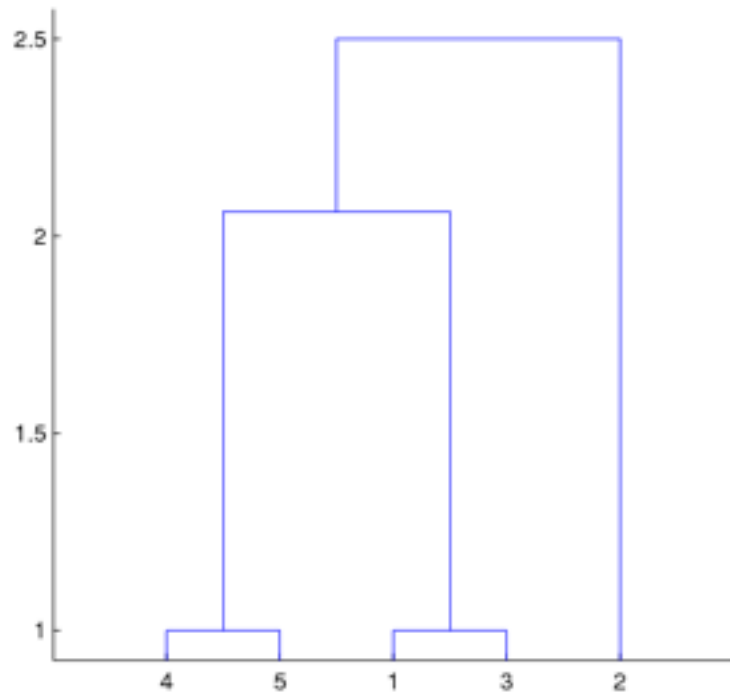
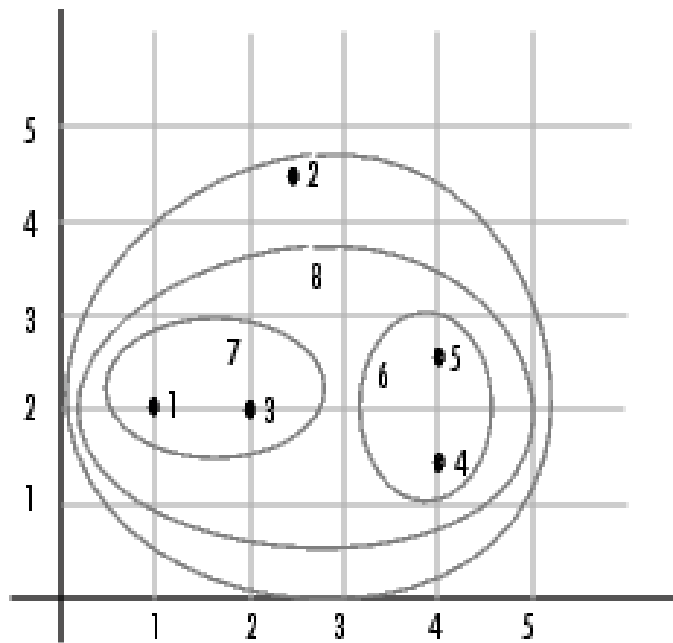
Introduction

- No longer a specification of K .
- **Alternatively:** user specified dissimilarity between disjoint groups of observations in the two groups.
- **Output:** Hierarchical representation of groups, the clusters at each level in the hierarchy are created by merging the groups at the level below, and adding an additional group.

```
z = linkage(Y)
```

```
z =
```

4.0000	5.0000	1.0000
1.0000	3.0000	1.0000
6.0000	7.0000	2.0616
2.0000	8.0000	2.5000



Introduction

Two Basic Forms result in $N-1$ levels of hierarchy.

- 1. Agglomerative** – bottom up, recursively merge a selected pair of clusters into a single cluster. Produces a grouping at the next higher level with one less cluster. Merging based on dissimilarity.
- 2. Divisive** – top down, at each level recursively split one of the existing clusters at that level into two new clusters. The split is chosen to maximize the between group dissimilarity.

Introduction

Input:

The entire data set. In high dimensional settings the data has to be filtered. Common filtering techniques: multivariate or univariate results, variation.

Output:

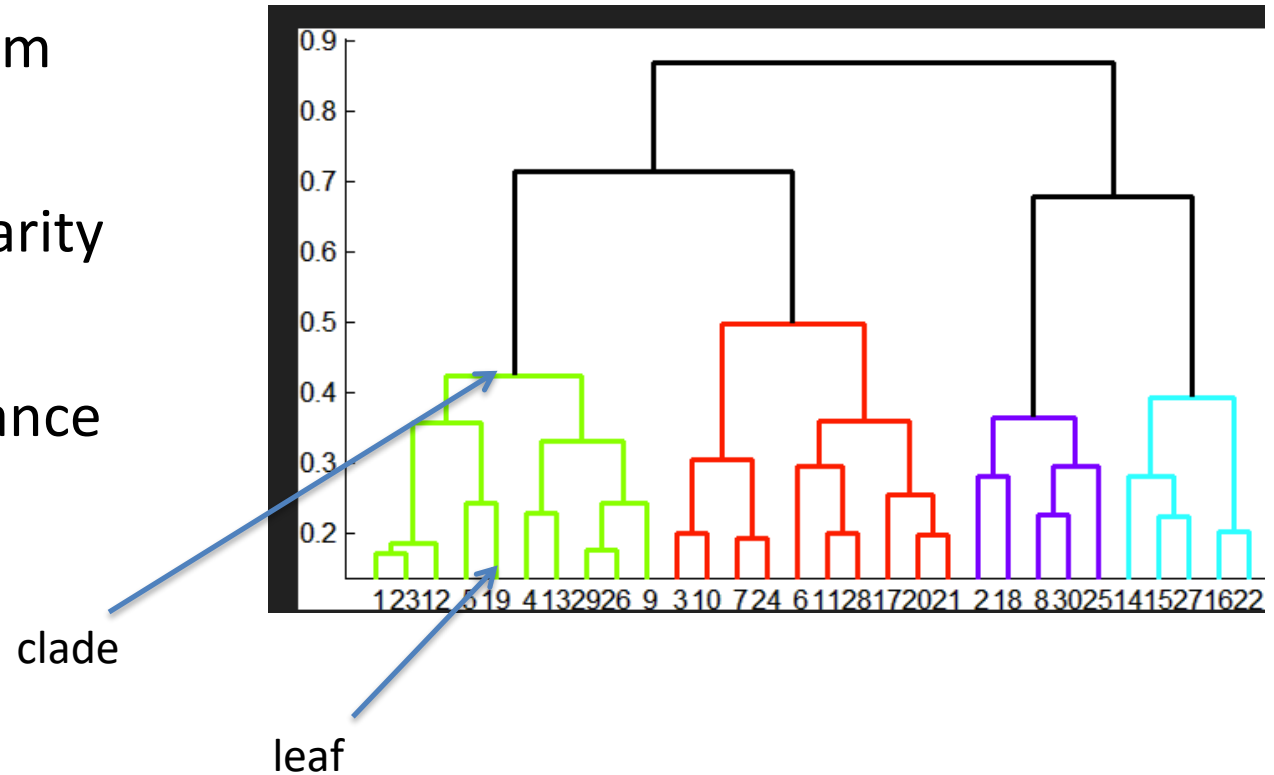
Similar in spirit to a binary tree

- The root node represents the entire input.
- The N terminal nodes each represent a single observation.
- Each non-terminal node (parent) has two children.

Dendrogram

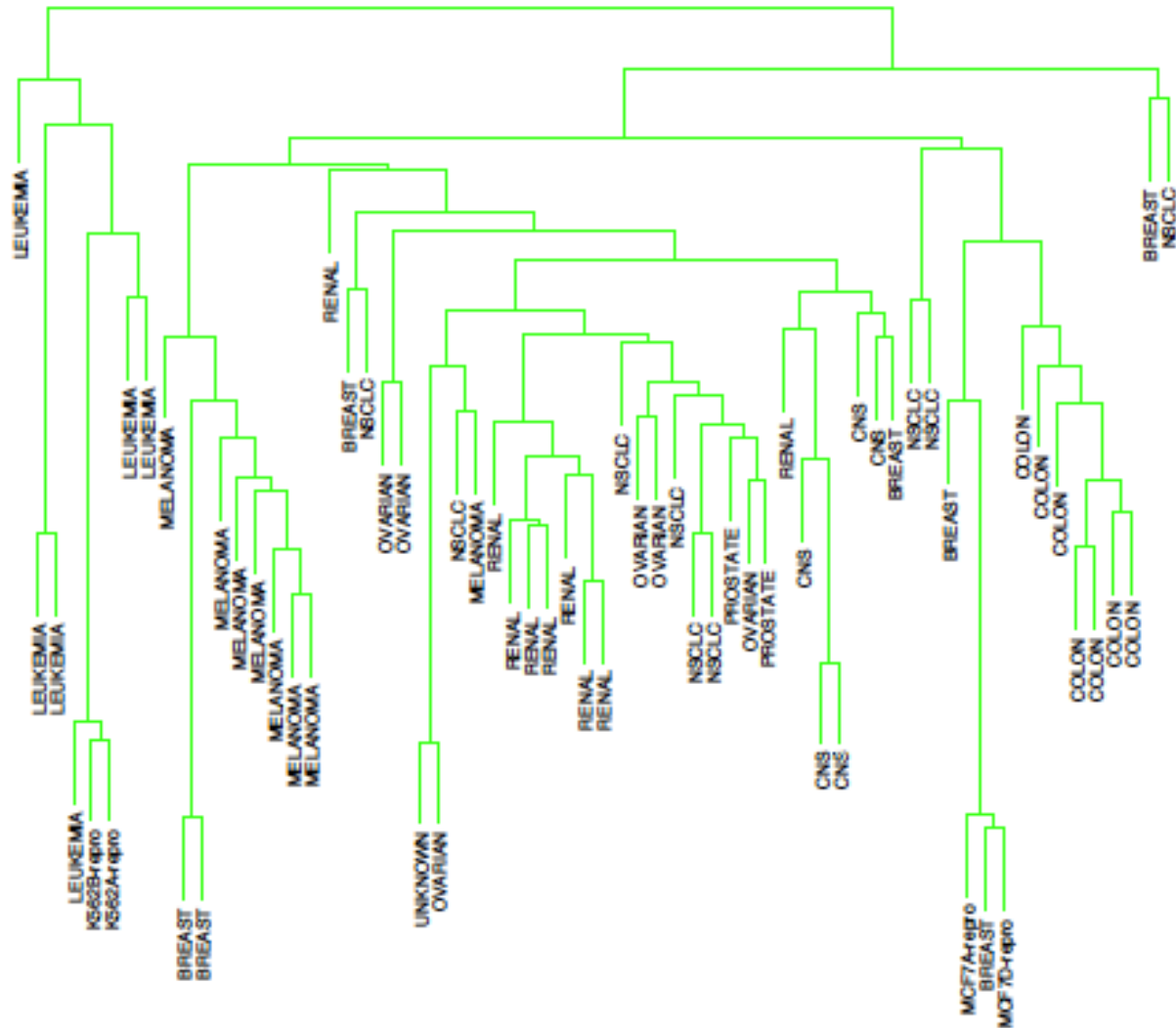
Dendrogram

- a branching diagram that represents the relationships of similarity among variables.
- Monotonic in distance



Dendrogram

Microarray Example (human tumor samples)



Clustering IV

When to stop

At some point, the user has to decide when to “cut” the tree.

- Gap statistic
- Dynamic Tree Cut
- Search for big jumps in distance, adding clusters decreases the clustering quality.
- Select a K, or a pre-specified cut level.
- Cophenetic Correlation

Defining clusters from a hierarchical cluster tree: the Dynamic Tree Cut package for R

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Agglomerative Clustering

Main Idea:

- N-1 Steps
- At each step, merge the two (least dissimilar/closest) into a single cluster.

More Formally:

- Let G and H represent two groups, and their dissimilarity is $d(G, H)$ computed from the set of pairwise observation dissimilarities $d_{ii'}$ where one member of the pair i is in G and the other i' is in H .

Agglomerative Clustering

Different Flavors (and others):

1. **Single Linkage:** takes the intergroup dissimilarity to be that of the closest (least dissimilar) pair:

$$d_{SL}(G, H) = \min_{\substack{i \in G \\ i' \in H}} d_{ii'}.$$

1. **Complete linkage:** takes the intergroup dissimilarity to be that of the furthest (most dissimilar) pair:

$$d_{CL}(G, H) = \max_{\substack{i \in G \\ i' \in H}} d_{ii'}.$$

2. **Group average:** uses the average dissimilarity between groups:

$$d_{GA}(G, H) = \frac{1}{N_G N_H} \sum_{i \in G} \sum_{i' \in H} d_{ii'}.$$

Agglomerative Clustering

- If the clusters are well-separated and compact, all metrics will unveil them.
- “compact” – the observations within them are relatively close together (small dissimilarities) as compared with the observations within different clusters.

**Single linkage may not yield compact groups.*

**Chaining – clusters may be forced together due to single elements being close, even though other elements in the cluster may be distant from each other.*

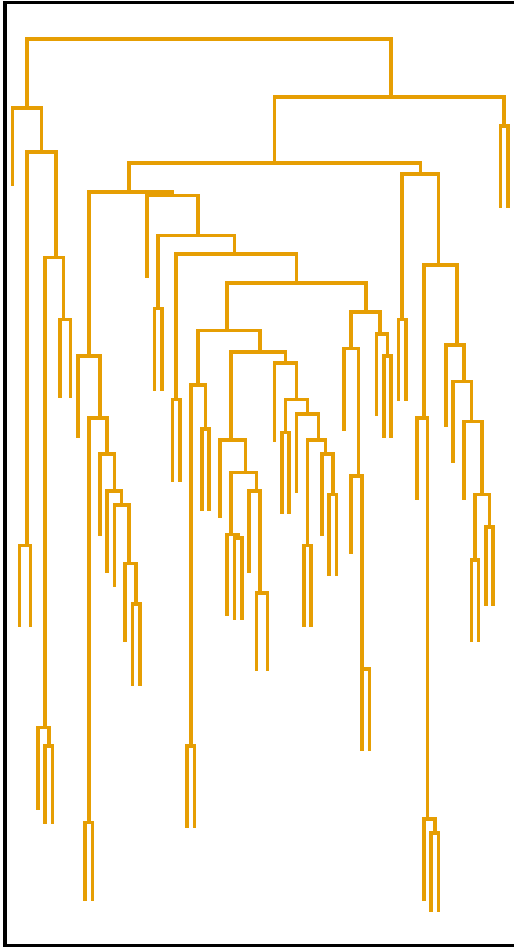
Can be thought of in terms of cluster diameter, $D_G = \max_{\substack{i \in G \\ i' \in G}} d_{ii'}$, which can be large in the single linkage case.

Agglomerative Clustering

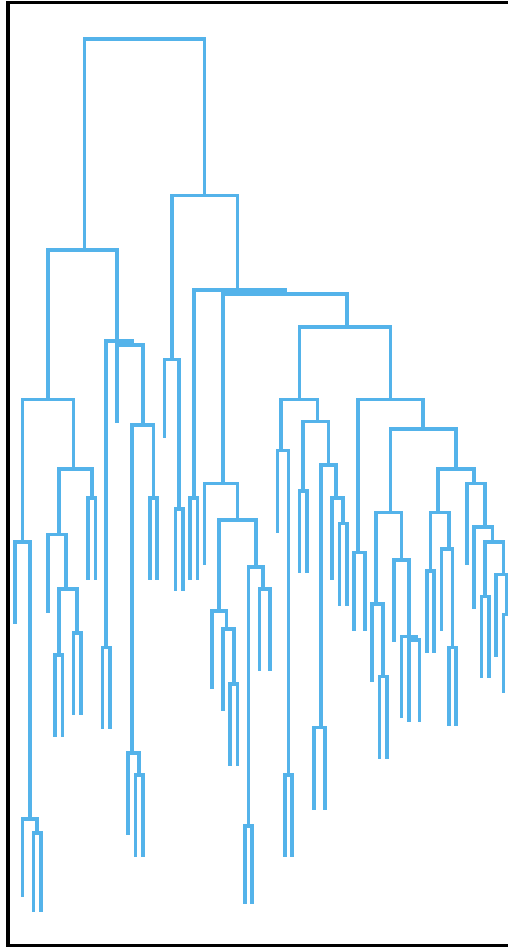
- Complete linkage has the opposite effect, two groups are considered close only if all the observations in their union are relatively close, producing a small $D_G = \max_{\substack{\hat{i} \in G \\ \hat{i}' \in G}} d_{\hat{i}\hat{i}'}$.
- Group average represents a compromise between single and complete linkage. Attempts to produce relatively compact clusters that are relatively far apart.

Example: Human Microarray

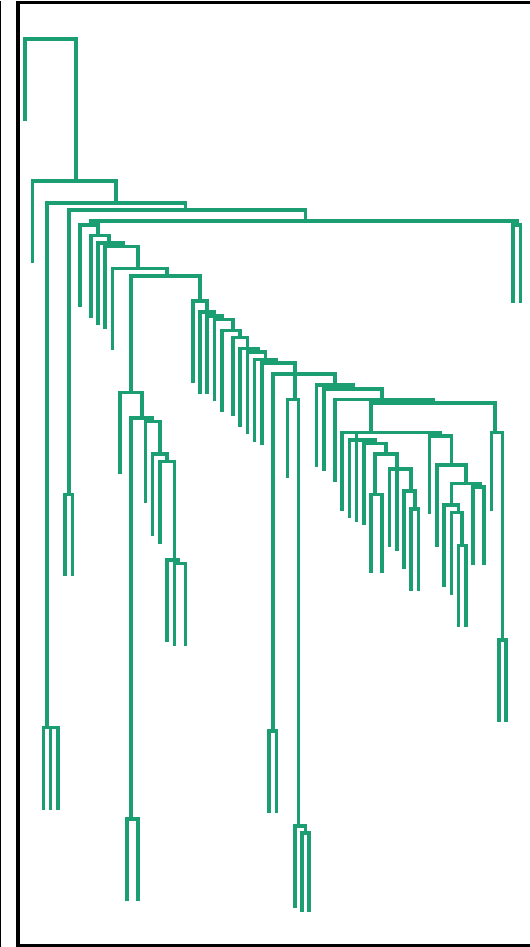
Average Linkage



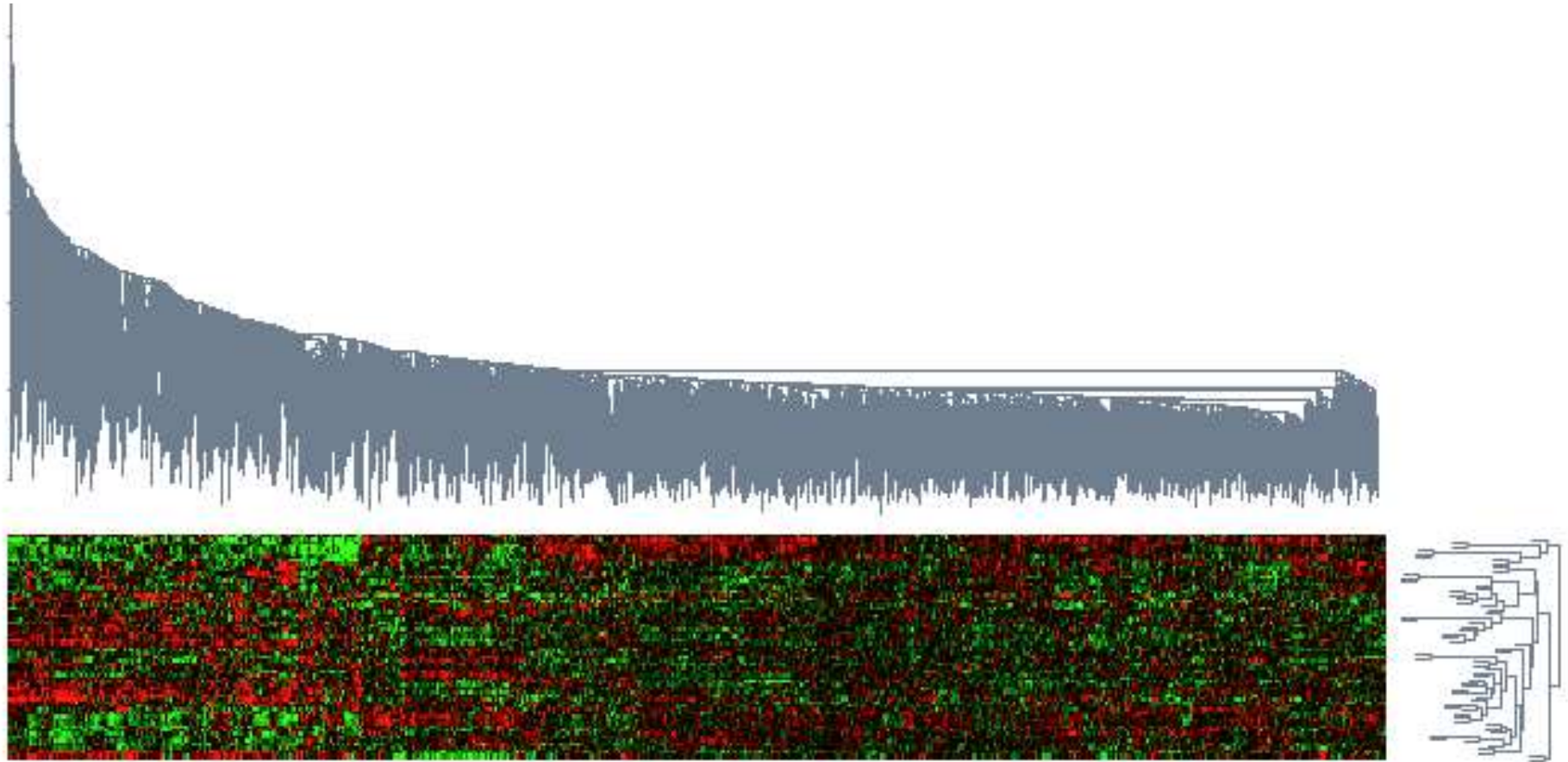
Complete Linkage



Single Linkage



Example: Human Microarray



Clustering IV